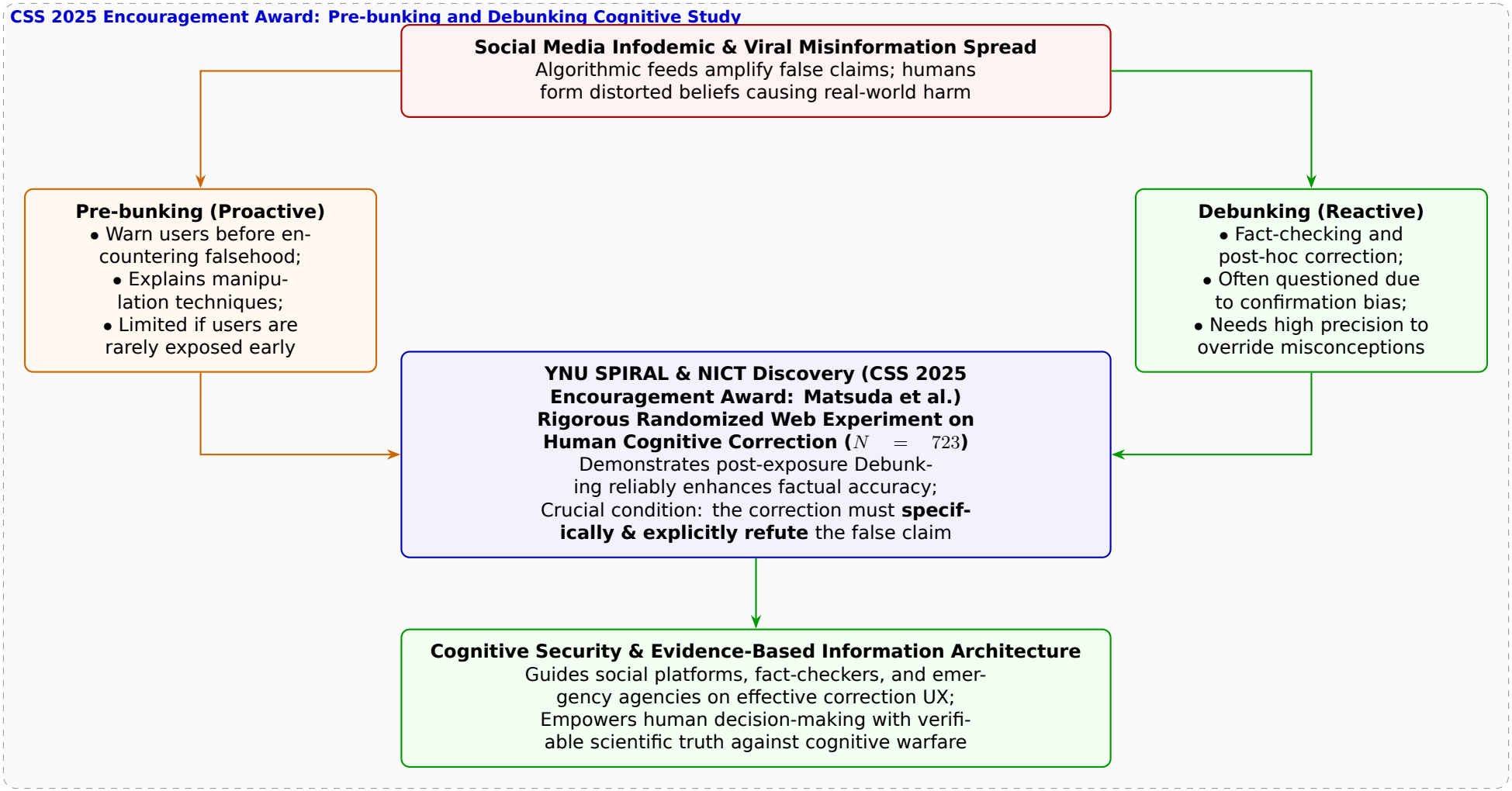

CSS 2025 Encouragement Award: Pre-bunking and Debunking Cognitive Study
Social Media Infodemic & Viral Misinformation Spread
Algorithmic feeds amplify false claims; humans form distorted beliefs causing real-world harm
Pre-bunking (Proactive)
• Warn users before encountering falsehood;
• Explains manipulation techniques;
• Limited if users are rarely exposed early
Debunking (Reactive)
• Fact-checking and post-hoc correction;
• Often questioned due to confirmation bias;
• Needs high precision to override misconceptions
YNU SPIRAL & NICT Discovery (CSS 2025 Encouragement Award: Matsuda et al.)
Rigorous Randomized Web Experiment on Human Cognitive Correction ($N = 723$)
Demonstrates post-exposure Debunking reliably enhances factual accuracy;
Crucial condition: the correction must specifically & explicitly refute the false claim
Cognitive Security & Evidence-Based Information Architecture
Guides social platforms, fact-checkers, and emergency agencies on effective correction UX;
Empowers human decision-making with verifiable scientific truth against cognitive warfare